End-to-End Method With Transformer for 3-D Detection of Oil Tank From Single SAR Image

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***Abstract*— In recent years, deep learning has been successfully applied in the field of synthetic aperture radar (SAR) image object detection. However, unlike ships and tanks targets, the oil tank targets in SAR image are usually dense and compact with more overlaps and discrete scattering centers, which greatly increases the difficulty of extracting location and structural parameters. Most of the existing methods transfer the methods suitable for natural image to the SAR image field, without considering the unique characteristics of SAR image. Therefore, in this article, we propose an improved model based on the end-to-end transformer network, which is the first model introducing transformer network to 3-D detection of oil tank targets from single SAR image. We input the incidence angle into the transformer model as a priori token. Then, we propose a feature description operator (FDO) based on the scattering centers that are used as an aid to improve the precision of predictions. In addition, we also propose a cylinder IOU as a more suitable evaluation metric for 3-D detection of oil tank. Finally, we evaluate our model on an SAR image dataset that contains SAR images from RADARSAT-2, TerraSAR-X, and GF-3 with different incidence angles. Our experiments demonstrate that our proposed model achieves the AP of 77.6% compared with 60.8% of baseline, which proves the effectiveness of the introduction of observation conditions, cylinder IOU loss (CI Loss), and the FDO based on the scattering centers in our model and is appealing for 3-D detection of oil tank.**

***Index Terms*— 3-D detection of oil tank, cylinder intersection over union (IOU) (cylinder IOU), feature description operator (FDO), priori token, single SAR image, transformer.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging sensor that can penetrate clouds, rain, snow, and smoke, and has all-day, all-weather imaging, and observation capabilities. It has been widely used in plenty of civilian fields due to its unique imaging mechanism [1]–[3]. Unlike optical sensors, SAR sensors can acquire images day and night and are not affected by clouds, fog, or smog. Due to the imaging mechanism, SAR images contain both azimuth and range information, which reflects the 3-D information of the imaging scene. Thus, the SAR images can provide more information than optical sensors under specific observational environmental conditions such as night and rainy weather and are more suitable for some typical object detection, such as building, ship, tank, and oil tank.

Oil tanks, most of which are built as cylinders and located above the ground, are important targets used by various countries to store crude oil. The 3-D detection of oil tanks (location, radius, and height) can be exploited to determine the maximum capacity of a given oil tank, which is important for both economies. Due to the unique characteristic of SAR, the oil tank detection and interpretation technology in SAR images has been paid attention to continuous development [4]–[6].

Remote sensing images have their own characteristics. SAR images are different from optical remote sensing images and also have their own characteristics. Therefore, we believe that we should combine the unique characteristics of SAR images to find a suitable method for solving SAR image problems using deep learning. In recent years, researchers have proposed some methods for detecting oil tanks using optical images. However, when applied them to SAR images, the performance is decreased because SAR images have their own particularities, such as side-view imaging, top and bottom inverted, too many scattering centers, and coherent speckle effects. These particularities will seriously affect image quality and interfere with subsequent detection. In most SAR images, only amplitude information is provided, which is the reflection of target electromagnetic scattering. The scattering of target is complicated with angle and frequency, and it exists in SAR image in the form of scattering centers. Coupled with some particularities of SAR image, SAR image application is challenging. This means that the boundaries of the oil tank cannot always be preserved and may result in accuracy reduced. The existing methods have been applied for oil tank detection from SAR images that can be divided into three categories. The first category is the method of fusing optical image and SAR image. Wang *et al.* [7] extracted bright regions from the SAR image to obtain candidate regions of the oil tank first and then performed shape detection on these corresponding candidate regions on the optical image before finally determining whether they are oil tanks. The

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second category is based on SAR image imaging geometry and scattering theory. Xu *et al.* [8] proposed a method for detecting oil tanks that combines circular shadows and highlight arcs. They detected circular shadows first, then found strong scattering points near the shadow area, and finally moved the detected circular edge to its corresponding strong scattering point. Back *et al.* [9] presented a method of using the detection of scattering points to analyze the oil storage trend of oil tanks. The method used KOMPSAT-5 SAR images and provided accurate oil storage status without manual intervention. Guida *et al.* [10] presented a method of exploiting the imaging geometry as well as its radiometry to estimate the height and diameter of cylindrical oil storage tanks with fixed roofs from SAR images. The third category is based on saliency analysis. Zhang *et al.* [11]–[14] presented several methods of exploiting saliency analysis to oil tank detection. The important limitation of these presented approaches is that the detection speed and accuracy cannot support the storage monitoring of oil tanks in long-term and large-scale scenarios. Most of these methods need to design features manually, and the scope of application of these methods is limited.

Except for oil tanks, the 2-D detection or 3-D reconstruction of other typical targets from SAR images have developed for a long time and achieved good results. Some researchers presented the fusion of SAR and optical features for 3-D reconstruction of buildings in SAR images [15], [16]. Traditional methods of extracting 3-D information of typical targets on SAR images include using overlay detection [17], an SAR simulator [18], GIS data [19], saliency analysis [20], KK distribution [21], density-censoring operation [22], image segmentation framework [23], statistical polygonal snakes [24], and geometrical structure prediction [25] and match the parameterized object model, the observed single intensity SAR image data [26], and so on. Although these methods have achieved their goals in their respective application scenarios through combining the imaging principles and unique features of SAR images, they can only be applied to small scenes or even only extract a single object and require complex manual design features. These shortcomings greatly limit the wide application of these methods. With the success of deep learning in the field of computer vision, some researchers hope to introduce deep learning in the SAR field to solve the above problems. The methods of deep learning include artificial neural network and support vector machine [27], dense attention pyramid networks [28], attention module and angle regression [29], triple-unit-graph and feature constraint representation learning [30], transitive transfer learning-based anchor-free rotatable model [31], data augmentation and transfer learning [32], semisupervised learning [33], and so on for 2-D detection of ship and plane targets. Most of these methods, which are based on convolutional neural network (CNN), only transfer the natural image processing models to the SAR image field, without considering the unique characteristics of SAR image such as imaging geometry mechanism, incidence angle, scattering information, and hidden targets' 3-D information. At the same time, the oil tank targets in SAR image are usually composed of discrete discontinuous scattering centers, which are relatively close to each other, and there are many overlapping phenomena, which limit the result of detection because of the receptive field of the CNN. The results of using CNN to detect oil tank targets are not satisfied. We want to explore a method that can provide a larger receptive field for 3-D detection of oil tank targets on SAR images and combine the unique characteristics of SAR images to improve the accuracy of detection.

Recently, with the rapid rise of transformer networks in the field of computer vision, it has inspired us to consider the unique characteristics of SAR in a new way. Transformer is a landmark model proposed by Google in 2017 [34], and it is the key technology of the language AI revolution. The great innovation of transformer lies in the parallelized language processing and relies on the self-attention mechanism. It has been the mainstream model only in the field of natural language processing until Facebook researchers presented DETR [35], which is the first model applying transformer to the field of object detection in computer vision. The full name of DETR is detection transformer, which is an end-to-end object detection network based on transformer. Traditional methods, such as anchor-based methods, are essentially the classification of predefined dense anchors and the regression of frame coefficients. DETR treats object detection as a set prediction problem. Since the transformer is essentially a sequence conversion function, DETR can be regarded as a conversion process from an image sequence to a collection sequence. This set is actually a learnable position encoding, which is also called object queries or output positional encoding in this article and $query_embed$ in the code. Next, some researchers from Google presented the ViT model [36] for image recognition, which converts image into serialized data through cutting image into patches arranged in a certain order. The presentation of these two articles opened the door to the application of transformer in the CV field. Many transformer variants and applications were quickly proposed and reached state of the art (SOTA) in other field of CV, such as image classification (DeiT [37], CLIP [38]), object detection (deformable-DETR [39]), low shot learning (CT [40]), image colorization (ColTran [41]), pose estimation (TFPose [42], PRTR [43]), action recognition (ST-TR [44]), Super-resolution (TTSR [45]), multimodel learning (Oscar [46]), 3-D classification/segmentation (PT [47]), 3-D mesh reconstruction (METRO [48]), vision and language navigation (Chen *et al.* [49]), referring image segmentation (CMSA [50]), and video classification (Lee *et al.* [51]). The main changes are related to specific loss function choices, architectural modifications, different position embeddings, and variations in input data modalities [52]. In the field of computer vision, transformer's self-attention mechanism and sequential input can provide long-distance receptive fields, which is very suitable for extracting oil tank targets composed of discrete scattering centers in SAR images. Also, its input is in the form of a sequence, which helps us to add other prior information of the SAR image.

Inspired by these works about transformer and the above problems, this article explores a way to combine the unique characteristics of SAR image by using a transformer for 3-D detection of oil tank from single SAR image. We analyze

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the imaging mechanism and characteristics of the SAR image in detail. For the oil tank targets, we believe that a single SAR image contains enough 3-D information. The main contributions of this work are as follows.

1) We first introduce transformer networks for 3-D detection of oil tank targets from a single SAR image to solve the problems of past related works that are the limitation of application scenarios.
2) The past applications of deep learning in the field of object detection from SAR images did not consider the unique characteristics of SAR images. We input the incidence angle information as *a priori* information into the model and considered utilizing scattering information to improve the performance.
3) Our model is an end-to-end SAR image 3-D detection network, without additional postprocessing operations, for example, NMS.
4) We approximate the oil tank target as a standard cylinder and propose a cylinder IOU as a more precise evaluation metric for 3-D object detection.
5) We evaluate our model on SAR datasets that contain a large number of SAR images with different incidence angles and prove the effectiveness and superiority of our model.

The remainder of this article is organized as follows. Section II introduces in detail the overall architecture of the proposed model, imaging geometric analysis of oil tank in SAR image, and some submodules. Section III introduces the experiments and results. Section IV discusses the reason why our proposed model is effective and superior. Finally, Section V summarizes this work.

II. Methodology

In this section, we first introduce the overall architecture of the proposed model, and then, we introduce the detailed imaging geometric analysis of oil tank in SAR image, feature description operator (FDO), transformer network architecture, prior token (PT), cylinder IOU, and details of the loss function.

A. Model Architecture

In this article, our goal is to extract the 3-D structural parameters of oil tank target (x_c, y_c, r, d), where (x_c, y_c) are the two-dimensional coordinates of top circumference center of oil tank target and r is the outer circumference radius of oil tank target. We set h as the height of oil tank target, so the imaging length d of SAR image is $d = h * \tan(\theta)$, where θ is the incidence angle.

The overall architecture of the proposed model is shown in Fig. 1. It contains a CNN backbone, a multiscale encoder–decoder transformer, and multilayer perception (MLP) prediction head. When training, we input a batch of SAR images with shape of 512×512 into the model. The features of SAR image are first extracted through the CNN backbone, and then, the incidence angle of SAR image is encoded as *a priori* token. The input of the multiscale transformer network encoder is the flattened feature maps extracted from the CNN backbone and PT. We set the range direction of the SAR image as prior information. Next, the input of the multiscale transformer decoder is N object queries, and each layer of the decoder receives the output from the multiscale transformer encoder. The output of the decoder is the 3-D structure parameters of N objects (x_c, y_c, r, d) and the probabilities of whether the objects are the ground-truth targets. According to the range direction of the SAR image and 3-D structure parameters of the SAR image, we can obtain the relevant parameters of the FDO to improve the accuracy of the 3-D structure parameters. The proposed model is an end-to-end method. When predicting, the probabilities corresponding to the predicted objects are sorted, and the threshold is set to retain the objects whose probabilities exceed the threshold.

B. Imaging Geometric Analysis of Oil Tank in SAR Image and Feature Description Operator

Oil tank targets in different regions usually have a cylindrical structure. The types of oil tank roofs include fixed roofs and floating roofs. Unlike the regular circular shape of oil tank target in the optical image, the oil tank profile in the SAR image usually appears as lots of discrete scattering centers, which are closely related to the imaging mechanism of SAR.

In this section, we analyze the imaging model of oil tank target in SAR image in detail, give the multipath scattering prediction model of oil tank target, and establish the relationship between multipath scattering characteristics and 3-D structure parameters of oil tank and radar parameters. We believe that single SAR image contains enough 3-D information of oil tank target.

In this article, we approximate the oil tank target as a standard cylinder and analyze the floating oil tank roofs. Fig. 2 shows the synthetic aperture radar imaging model of oil tank target.

According to the literature [53], we use a physical optics method (PO) [54] and incremental length diffraction coefficients (ILDC) [55] theory to establish the scattering model of oil tank target. As shown in Fig. 2, when the oil tank target is irradiated by radar electromagnetic waves, the scattered field is mainly composed of the following parts: the sidewall of oil tank and the outer circumference edge of oil tank roofs ABC, the inner edge of circumference $A'B'C'$, and the bottom circumference of oil tank DEF.

At first, we analyze the scattering information of oil tank sidewall. Generally, the irradiated oil tank inner wall contributes less. We can replace the scattering information of oil tank inner wall with the scattering with the parabolic shape (the intersection of roof and cylinder) approximately. The outer wall of oil tank has a greater contribution to the scattered field. In the imaging model shown in Fig. 2, assuming that the electric field is horizontally polarized, the incidence electromagnetic wave vector $\mathbf{k}$ is $(\sin\theta, 0, \cos\theta)$, and according to the physical optics (PO) method, the scattered field E^s is

$$E^s = \frac{-jke^{jkR}}{2\pi R}\int_s \left[(\mathbf{e}_\perp \cdot \mathbf{e}_i)^2 \Gamma_\perp + \left(\mathbf{e}_\parallel \cdot \mathbf{e}_i\right)^2 \Gamma_\parallel\right](\mathbf{n}\cdot\mathbf{k}) \cdot e^{j2k\mathbf{r}\cdot\mathbf{k}}\, ds \tag{1}$$

where $\mathbf{n}$ is the local normal vector, $\mathbf{r}$ is the local direction vector, k is the wavenumber, $\Gamma_\perp$ and $\Gamma_\parallel$ represent the vertical

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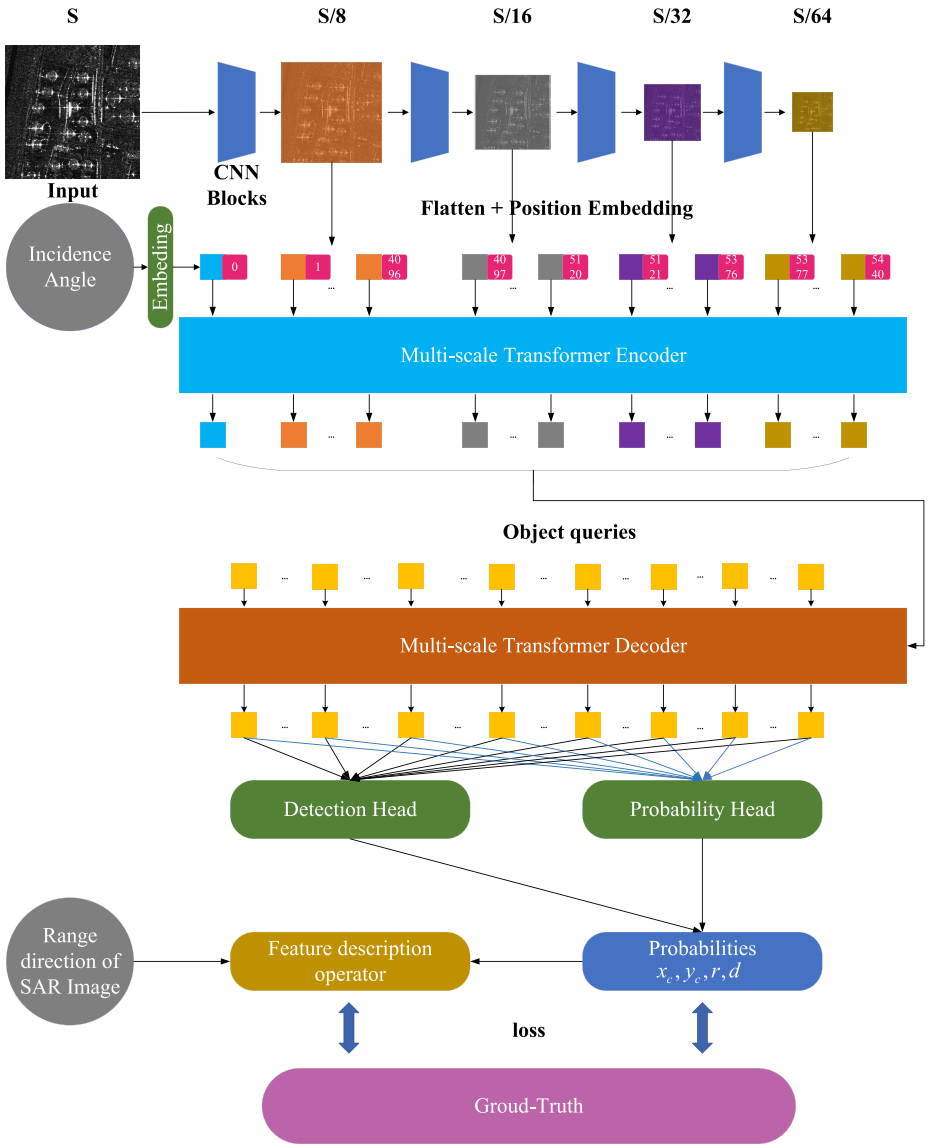


Fig. 1. Overall architecture of the proposed model. We set the size of the input image to S, and then, the input of multiscale transformer includes four kinds of feature maps with resolutions of $(S/8)$, $(S/16)$, $(S/32)$, and $(S/64)$, which are flattened and concated into a 1-D sequence. The pink boxes at the input of multiscale transformer encoder represent the position information coding of different tokens in the 1-D sequence. The FDO of each oil tank target, which is defined as the coordinates of two strong scattering points, is calculated from the 3-D structure parameters output by the model and the range direction of SAR image, where the range direction of SAR image represents the incident direction of electromagnetic waves of the SAR satellite and reflects the relative positional relationship of two strong scattering points.

polarization and horizontal polarization, respectively, $\mathbf{e}_\perp$ and $\mathbf{e}_\parallel$ represent the direction of vertical polarization and horizontal polarization, respectively, and s represents the area of electromagnetic wave irradiation. We set $f(\varphi)$ that is related to material of oil tank

$$f(\varphi) = \left[(\mathbf{e}_\perp \cdot \mathbf{e}_i)^2 \Gamma_\perp + \left(\mathbf{e}_\parallel \cdot \mathbf{e}_i\right)^2 \Gamma_\parallel\right](\mathbf{n} \cdot \mathbf{k}) \tag{2}$$

$$f(\varphi) = A(\varphi)\exp(jB(\varphi)) \tag{3}$$

where $A(\varphi)$ and $B(\varphi)$ represent amplitude and phase, respectively.

Thus, the scattered field can be written as

$$E^s = \frac{-jke^{jkR}}{2\pi R}\int_s A(\varphi)\mathbf{e}^{j[2k\mathbf{r}\cdot\mathbf{k}+B(\varphi)]}\,ds \tag{4}$$

where R is the distance between oil tank target and satellite.

Substituting formula (4) into formula (1), we have

$$\begin{aligned} E^s = a\frac{-jkE_0}{2\pi}\frac{\exp(-jkr)}{r}\sin\theta \\ \cdot\theta\int_{-\pi/2}^{\pi/2} A\left(\varphi'\right)\exp\left(jB\left(\varphi'\right)\right) \\ \cdot\exp 2jka\sin\theta\cos\varphi'\,d\varphi' \\ \cdot\int_{z(\varphi')}^{H}\exp 2jkz'\cos\theta\,dz' \end{aligned} \tag{5}$$

where θ is the direction vector in cylindrical coordinate system and a is the radius of oil tank bottom circle. Also

$$z' = \begin{cases} H - a\left(1+\cos\varphi'\right)\tan\theta, & \text{Inner wall} \\ 0, & \text{Outer wall.} \end{cases} \tag{6}$$

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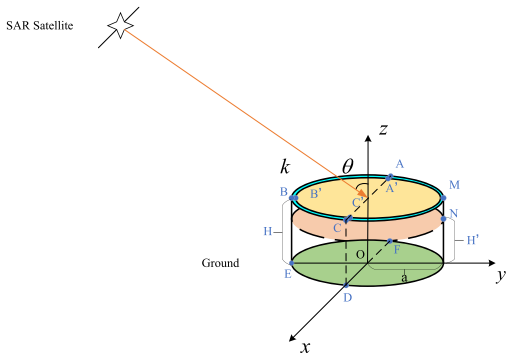


Fig. 2. SAR imaging model of one oil tank target. The yellow, orange, and green circles represent the top, roof, and bottom of oil tank, respectively. θ is the incidence angle and k is the incident vector. Plane XOY represents the ground plane and the axis z represents the height axis. Note that we set the incident vector k to be perpendicular to the x-axis, H' is the height of floating oil tank roof (orange circle) from the ground, and H is the height of oil tank top (yellow circle) from the ground.

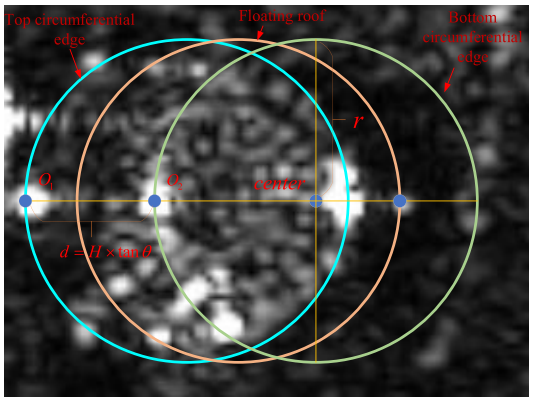


Fig. 3. Strong scattering information of oil tank target. The top circumferential edge is drawn as blue circle, the floating roof is drawn as orange circle, and the bottom circumferential edge is drawn as green circle. The bottom circle center, feature operator points O_1 and O_2, and structure parameters r and d ($H \times \tan\theta$) are also drawn on this figure.

According to formula (4), the characteristics of phase term is: $2k\mathbf{r}\cdot\mathbf{k} \gg B(\varphi)$, and the change of $A(\varphi)$ j is much smaller than the phase envelope. When $z'(\varphi) = 0$

$$E^s = \theta \frac{E^0 a \tan\theta}{4\pi} \frac{\exp(-jkr)}{r} I \cdot \left[\exp(2jkH\cos\theta) - 1\right] \quad (7)$$

where

$$I = \int_{\pi/2}^{3\pi/2} f(\varphi') \exp\left(2jka\sin\theta\cos\varphi'\right) d\varphi'. \quad (8)$$

According to the principle of stationary phase (POSP) [56], we have the stationary phase point $\varphi' = \pi$ and

$$I = -\left(\frac{\pi}{jka\sin\theta}\right)^{1/2} \exp(-2jka\sin\theta) A(\pi) \quad (9)$$

so the scattered field is

$$E^s = \theta \frac{E_0 a \tan\theta}{4\pi} \frac{\exp(-jkr)}{r} \left(\frac{\pi}{jka\sin\theta}\right)^{1/2} A(\pi) \cdot \exp(-j2ka\sin\theta) \cdot \left[\exp(2jkH\cos\theta) - 1\right]. \quad (10)$$

When $z'(\varphi) = H - a(1+\cos\varphi')\tan\theta$, the scattered field is

$$E^s = \theta \frac{E_0 a \tan\theta}{4\pi} \frac{\exp(-jkr)}{r} \left(\frac{\pi}{jka\sin\theta}\right)^{1/2} A(\pi) \cdot \exp(-j2ka\sin\theta) \exp(2jkH\cos\theta) \cdot \left\{-\left(\frac{\pi}{jk\cos\theta}\right)^{1/2} A(\pi) - \int_{-\pi/2}^{\pi/2} f(\varphi')\, d\varphi'\right\}. \quad (11)$$

According to formulas (10) and (11), we inferred the scattering model of oil tank target as follows; the outer wall of oil tank target will form two strong scattering centers at points B and E in Fig. 2, whose coordinates are $(0, -a, 0)$, $(0, -a, H)$, and the inner wall of oil tank will form a strong scattering center at point N $(0, a, H')$, which is the junction of the floating roof and the inner wall of oil tank. At point M $(0, a, H)$, it forms a scattering center with a smaller scattering intensity.

Next, we analyze the scattering model of circumferential edge of oil tank target. First, the edge of circle is modeled by line elements. According to incremental length diffraction coefficients (ILDC), in the backscattering, the scattered field of each line element is derived as

$$E^s = l \frac{\exp[j(kr - \pi/4)]}{\sqrt{2\pi r}} \exp(2jk\mathbf{r_c}\cdot\mathbf{k}) \cdot \sin c(kl\mathbf{p}\cdot\mathbf{k}) \overline{\overline{d}} \cdot E^i \quad (12)$$

where $\mathbf{p}$ is the line element direction vector, $\mathbf{r_c}$ is the line element center position vector, and $\overline{\overline{d}}$ is diffraction coefficient. At this time, the scatter calculation process at the corresponding positions of four points A, C, D, and F in Fig. 2 will appear singular.

According to the analysis of radar imaging model, in the SAR image, the strong scattering centers of oil tank target mainly include the following situations.

1) The first scattering of outer circumference belongs to the top of oil tank, and this position is closest to the satellite position, which is clearly contrast with the ground. In Fig. 2, it is defined as point B that presents a stronger scattering center.
2) Multiple scattering between the outer wall of oil tank and ground. Due to multiple scattering at this location, it is defined as point E that presents a stronger scattering center.
3) Multiple scattering between the floating roof and the inner wall of oil tank. It is defined as point N that presents a stronger scattering center. In addition, the inner top circumference, the top circumferential edges, the floating roof, and the bottom of the oil tank will also form weaker intensity scattering. Fig. 3 shows a schematic of the strong scattering centers of oil tank target.

According to the scattering model of oil tank target deduced above, we propose the FDO of oil tank target, which is defined as the position of the two strong scattering centers formed by the outer wall of oil tank target at points O_1 and O_2 in Fig. 3, which corresponds to points B and E in Fig. 2. These two strong scattering centers can better reflect the radius and

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height information of oil tank target in the SAR image. In the following, we will use the FDO to assist in improving the precision of the 3-D structure parameters extraction of oil tank target.

C. Revising Transformer, DETR, and Deformable-DETR

Deformable-DETR [39] is improved from DETR [35] that treats object detection problem as a set prediction problem to simplify the overall process of object detection. In the following, we will briefly introduce transformer, DETR, and deformable-DETR that are used in our proposed model.

Transformer [34] is a network architecture based on self-attention mechanism. It consists of several encoder layers and decoder layers. In each encoder layer, the input goes through four sublayers in turn: embedding, positional encoding, multihead attention, and position-wise feedforward. Residual connection is made between each sublayer. In each decoder layer, a masked multihead attention sublayer is added compared with encoder layer. Finally, the input passes through the linear and softmax layers to output the probability. In the following, we briefly review multihead attention, which is the core of transformer. Given a query element and a set of key elements, the multihead attention module adaptively aggregates the key contents according to the attention weights that measure the compatibility of query–key pairs. The outputs of different attention heads are linearly aggregated through learnable weights so that the model focuses on content from different representation subspaces and different locations. The multihead attention feature is calculated by

$$\text{MultiHeadAttn}\big(z_q, x\big) = \sum_{m=1}^{M} W_m \left[\sum_{k\in\Omega_k} A_{mpk} \cdot W'_m x_k\right] \quad (13)$$

where $q \in \Omega_q$ indexes a query element with representation feature, $z_q \in \mathbb{R}^C$, $k \in \Omega_k$ indexes a key element with representation feature, $x_k \in \mathbb{R}^C$, C is the feature dimension, Ω_q and Ω_k specify the set of query and key elements, m indexes the attention head, and $W'_m \in \mathbb{R}^{C_v \times C}$ and $W_m \in \mathbb{R}^{C \times C_v}$ are learnable weights. $A_{mpk} \propto \exp\{((z_q^T U_m^T V_m x_k)/(C_v)^{1/2})\}$ are the attention weights that are normalized as $\sum_{k\in\Omega_K} A_{mpk} = 1$, and U_M and V_m are learnable weights.

DETR, an end-to-end object detection network based on transformer treats object detection as a set prediction problem, which can be regarded as a conversion process from an image sequence to a collection sequence. The architecture of DETR includes CNN backbone, transformer encoder, transformer decoder, and output layers. CNN backbone is to extract the feature map $x \in \mathbb{R}^{C\times H\times W}$ with one specific resolution. For the DETR encoder, the query elements and key elements are pixels in the feature map x with position embeddings that provide position information of each pixel in the feature map. The attention mechanism in encoder is the self-attention mechanism and the attention mechanism in decoder includes self-attention mechanism and cross-attention mechanism. In the self-attention module, object queries influence each other to capture the relationship between them. In the cross-attention module, the object queries extract features from the encoder feature map, the query element belongs to the object query, and the key element belongs to the output feature map from the encoder. For the output layers, a three-layer feedforward neural network (FFN) and a linear layer are added to the object query feature (produced by the decoder) as the detection head, and another FFN layer outputs the class of the object boxes (class or no object).

Although DETR has an interesting design and good performance, DETR also has its own problems. First, compared with the existing detection methods, it needs longer training time to converge. On the COCO dataset, it needs 500 epochs to converge, which is 10–20 times faster than the faster r-cnn. Second, DETR has poor performance in the detection of small objects. Existing detectors usually use multiscale features. Small objects are usually detected on high-resolution feature maps, while DETR does not use multiscale features to detect, and high-resolution feature maps will add unacceptable computational complexity to DETR.

Therefore, in order to solve the above problems of DETR, deformable-DETR combines the advantages of deformable sparse spatial sampling and the relationship modeling ability of transformer and proposes a multiscale deformable attention module on the basis of DETR architecture. For each query, DETR pays attention to all spatial positions (all positions are used as keys), and deformable-DETR only pays attention to the more meaningful positions that the network considers to contain more local information (a small and fixed number of positions are used as keys) to alleviate the problem of a lot of calculations that come from the large band of feature maps. On the basis of multihead attention, the deformable attention feature is calculated by

$$\text{DeformAttn}\big(z_q, p_q, x\big) = \sum_{m=1}^{M} W_m \cdot \left[\sum_{k=1}^{K} A_{mpk} \cdot W'_m x\big(p_q + \delta p_{mpk}\big)\right] \quad (14)$$

where p_q is a 2-d reference point, k indexes the sampled keys, and K is the total sampled keynumber ($K \ll HW$). δp_{mpk} denotes the sampling offset and attention weight of the kth sampling point in the mth attention head. Extended for multiscale feature maps, if $x^{l}{}_{l=1}^{L}$ is the input multiscale feature map, the multiscale deformable attention module is applied as

$$\text{MSDeformAttn}\Big(z_q, \hat{p}_q, x^{l}{}_{l=1}^{L}\Big) = \sum_{m=1}^{M} W_m \left[\sum_{l=1}^{L}\sum_{k=1}^{K} A_{mlpk} \cdot W'_m x^{l}\big(\phi_l \hat{p}_q + \delta p_{mlpk}\big)\right] \quad (15)$$

where l indexes the input feature level.

The input and output of deformable-DETR encoder layers are multiscale feature maps of the same resolution. In order to verify which feature layer each query pixel is in, in addition to the location feature vector, deformable-DETR also adds a scale-level feature vector e_l to the feature representation. Deformable-DETR decoder layers replace each cross-attention module with a multiscale deformable attention module and keep the self-attention module unchanged. The output layers are the same as DETR.

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D. Prior Token and Input of Multiscale Transformer Network

The input of standard transformer network is 1-D token embedding. CNN has achieved great success in the field of image feature extraction, which has translation invariance and local perception of image feature extraction. However, due to the limitation of the receptive field of the convolution kernel, it is limited in processing the global reasoning. The transformer network can achieve long-distance prediction and global reasoning due to its own self-attention mechanism. The limitation is that it lacks more refined local features if we apply transformer to the computer vision field directly, but we can complement each other through the combination of the two networks to deal with. According to the literature [39], in order to process two-dimensional feature maps after using CNN's multiscale feature extraction, the input of multiscale transformer is set to the multiscale flattened feature maps extracted by the CNN backbone, with position embedding and level embedding that are added to provide location information and feature map scale information. In this article, the CNN backbone we used is Resnet50, and the input image is downsampled by CNN into feature maps of four different resolutions that are input to the multiscale transformer. As shown in Fig. 1, we set the size of the input image to S, and then, the four resolutions that input to the transformer network are $(S/8)$, $(S/16)$, $(S/32)$, and $(S/64)$.

Considering that unlike natural images, the characteristics of SAR images are greatly affected by the observation conditions of radar satellites. Under different observation conditions, the same target will present completely different forms, and the incidence angle is the most important among observation conditions. We consider adding the incidence angle prior to the input of the model to improve performance. The transformer network can obtain good results for the global reasoning of the relationship between the input 1-D tokens. Therefore, we believe that the prior information of the incidence angle is used as the input of the multiscale transformer encoder, and the self-attention mechanism of the network can be used to learn the relationship between the incidence angle and SAR image features more accurately. Inspired by the vision transformer (ViT) model [36], we use the incidence angle as a special PT and set position embedding to 0 before inputting into the multiscale transformer network. The embedding corresponding to the PT is randomly initialized during training and then obtained through training, so the input sequence length of the multiscale transformer network becomes N+1, where N is the total dimension of the multiscale feature maps after flattening and cascading. The multiscale transformer network is shown in Fig. 4.

E. Cylinder IOU

In canonical object detection, intersection over union (IOU) is the most popular evaluation metric, which is used to evaluate the similarity between predicted and ground-truth objects. IOU is defined as the ratio of the intersection area to the union area. In this article, we regard the oil tank target as a standard cylinder, so the extracted 3-D structural parameters are the structural parameters of the cylinder. Different from other 3-D detection and extraction of cuboid bounding box, the use of cuboid IOU as evaluation metric can no longer reflect the precision of the predicted 3-D structure parameters. Especially when the cuboid bounding box intersects, the two cylinders are not necessary for an intersecting state. Therefore, we propose cylinder IOU as a more accurate evaluation metric. The definition of cylinder IOU is introduced as follows.

Similar to the calculation of the circle IOU [57], we set A and B as two cylinders that are prepared to compute intersection over union as in Fig. 5. The definition of the cylinder IOU is as follows:

$$\text{cylinder IOU} = \frac{\text{Intersection volume}(A, B)}{\text{Union volume}(A, B)}. \tag{16}$$

The bottom circle center coordinates of cylinders A and B are defined as (x_A, y_A) and (x_B, y_B), respectively, and the bottom circle radii of cylinders A and B are defined as r_A and r_B, respectively, and the heights of A and B are defined as h_A and h_B, respectively. Thus, Intersection volume(A, B) is defined as

$$\text{Intersection volume}(A, B) = \mathcal{S}_{\text{intersection}} \times \min(h_A, h_B) \tag{17}$$

where $\mathcal{S}_{\text{intersection}}$ is the intersection area between the bottom circle of cylinders A and B. Union volume(A, B) is defined as

$$\begin{aligned}\text{Union volume}(A, B) = \pi\left(r_A^2 \times h_A + r_B^2 \times h_A\right)\\ -\text{Intersection volume}(A, B).\end{aligned} \tag{18}$$

In the following, we give the derivation process of $\mathcal{S}_{\text{intersection}}$. Consider the three situations shown in Fig. 6: two circles are separated, contained, and partially overlap situations. When two circles are separated, the intersection area between two circles is zero. When two circles are contained, the intersection area between two circles is the area of the smaller circle. Thus, we focus on partially overlapped situations that are more complicated.

The distance between the center coordinates d is defined as

$$d = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2} \tag{19}$$

and partially overlapped situations can be divided into two situations as in Fig. 7. Assume that $r_B \leqslant r_A$, so θ_A and θ_B can be defined as, respectively

$$\theta_A = \arccos\frac{r_A^2 + d^2 - r_B^2}{2r_A d} \tag{20}$$

$$\theta_B = \arccos\frac{r_B^2 + d^2 - r_A^2}{2r_B d}. \tag{21}$$

In the first situation, $\theta_B \leqslant (\pi/2)$, we can get $\mathcal{S}_{\text{intersection}}$ as follows:

$$\begin{aligned}\mathcal{S}_{\text{intersection}} &= \mathcal{S}_{\text{sectorA}} + \mathcal{S}_{\text{sectorB}} - \mathcal{S}_{\text{triangleA}} - \mathcal{S}_{\text{triangleB}}\\ &= r_A^2 \times \theta_A + r_B^2 \times \theta_B - r_A \sin\theta_A \times r_A \cos\theta_A\\ &\quad - r_B \sin\theta_B \times r_B \cos\theta_B\\ &= r_A^2(\theta_A - \sin\theta_A \cos\theta_A) + r_B^2(\theta_B - \sin\theta_B \cos\theta_B).\end{aligned} \tag{22}$$

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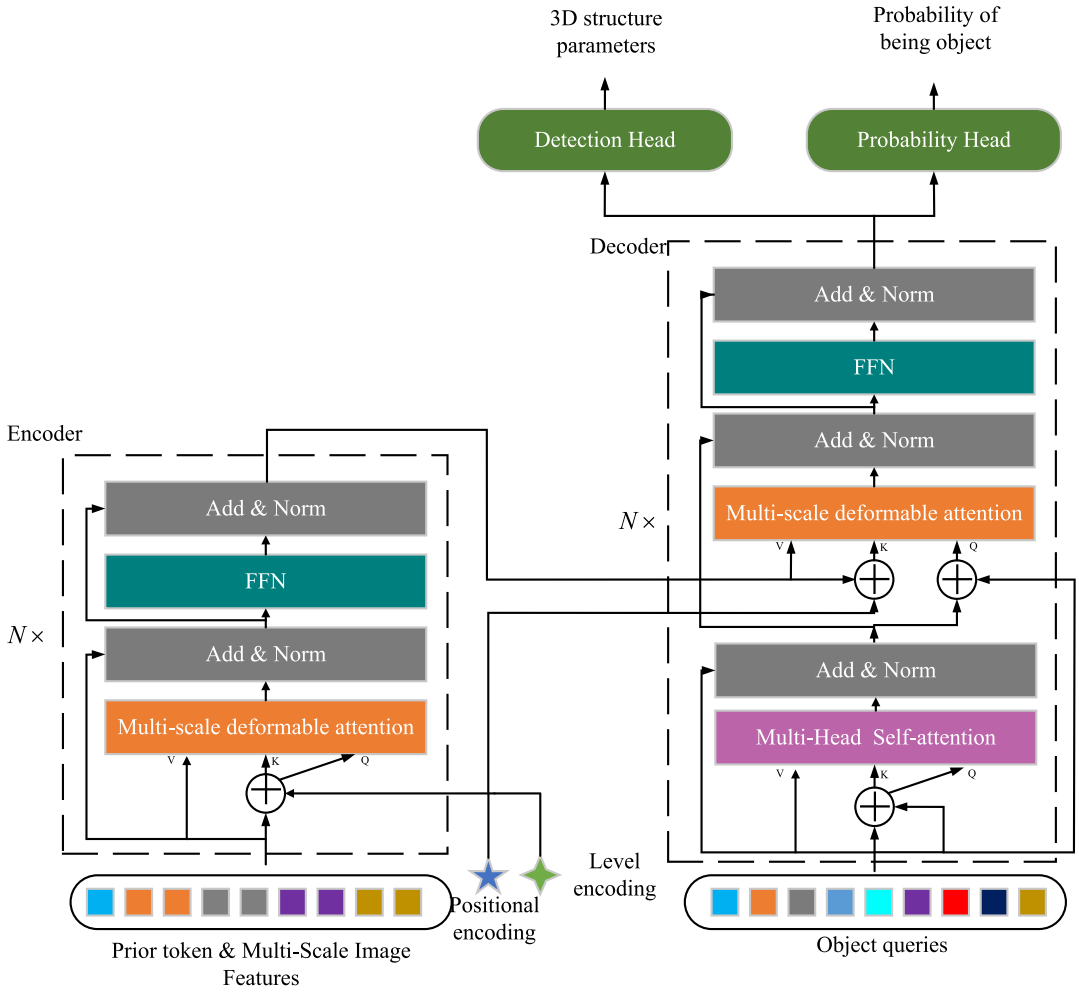


Fig. 4. Multiscale transformer network of our proposed model.

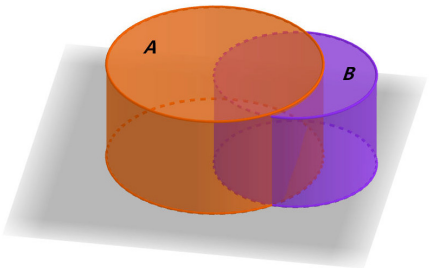


Fig. 5. Cylinders that are prepared to compute intersection over union. Note that cylinders A and B, which are located on the ground, represent the predicted and ground-truth objects, respectively.

In the second situation, $\theta_A \geqslant (\pi/2)$, we can get $\mathcal{S}_{\text{intersection}}$ as follows:

$$\begin{aligned}\mathcal{S}_{\text{intersection}} &= \mathcal{S}_{\text{sectorA}} + \mathcal{S}_{\text{sectorB}} + \mathcal{S}_{\text{triangle-B}} - \mathcal{S}_{\text{triangle-A}} \\ &= r_A^2 \times \theta_A + r_B^2 \times \theta_B + r_B \sin(\pi - \theta_B) \\ &\quad \times r_B \cos(\pi - \theta_B) - r_A \sin\theta_A \times r_A \cos\theta_A \\ &= r_A^2 \times \theta_A + r_B^2 \times \theta_B - r_B \sin\theta_B \times r_B \cos\theta_B \\ &\quad - r_A \sin\theta_A \times r_A \cos\theta_A \\ &= r_A^2(\theta_A - \sin\theta_A \cos\theta_A) + r_B^2(\theta_B - \sin\theta_B \cos\theta_B)\end{aligned} \tag{23}$$

so $\mathcal{S}_{\text{intersection}}$ can be calculated as follows:

$$\begin{aligned}\mathcal{S}_{\text{intersection}} &= r_A^2(\theta_A - \sin\theta_A \cos\theta_A) \\ &\quad + r_B^2(\theta_B - \sin\theta_B \cos\theta_B).\end{aligned} \tag{24}$$

F. Loss Function

The output of the proposed model is a set of N predicted objects $\hat{\mathcal{T}} = \{\hat{t}_i\}_{i=1}^{N}$, where N is set to larger than the typical number of single SAR image. Each predicted object consists of four parts: object and nonobject probability $\hat{\mathcal{P}} = (\hat{p}_{\text{Object}}, \hat{p}_{\text{NonObject}}) \in [0,1]^2$, 3-D structure parameters $\hat{\mathcal{B}} = (\hat{x}_c, \hat{y}_c, \hat{r}, \hat{d}) \in [0,1]^4$, range direction probability $\hat{\mathcal{D}} = (\hat{d}_{\text{left}}, \hat{d}_{\text{right}}) \in [0,1]^2$, and FDO parameter $\hat{\mathcal{O}} = (\Delta\hat{x}_{O_1}, \Delta\hat{y}_{O_1}, \Delta\hat{x}_{O_2}, \Delta\hat{y}_{O_2}) \in [-0.5, 0.5]^4$.

Since the number of predicted objects is much larger than the number of ground-truth objects, similar to the literature [35], we use the Hungarian algorithm [58] to obtain an

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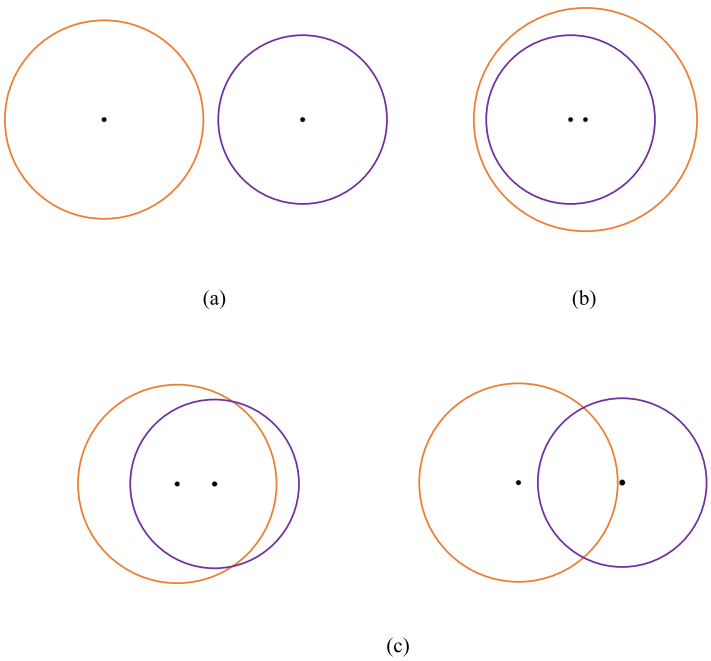


Fig. 6. Three situations when calculating $\mathcal{S}_{\text{intersection}}$. (a) Situation that two circles are separated. (b) Situation that two circles are contained. (c) Situation that two circles are partially overlapped.

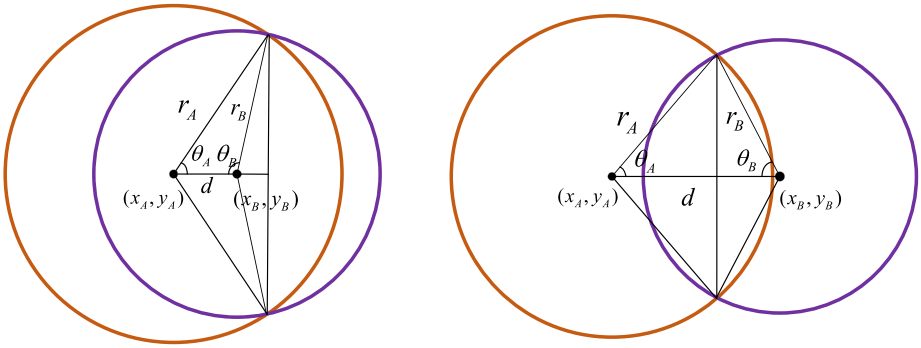


Fig. 7. Two situations that two circles are partially overlapped.

optimal bipartite matching between predicted and ground-truth objects, searching for a permutation of N elements $\sigma \in \mathcal{G}_N$ with the lowest cost

$$\hat{\sigma} = \arg\min \sum_{i}^{N} \mathcal{L}_{\text{match}}\left(t_i, \hat{t}_{\sigma(i)}\right) \tag{25}$$

where t presents the ground-truth objects that are padded to length of N with nonobject and $\mathcal{L}_{\text{match}}(t_i, \hat{t}_{\sigma(i)})$ is a pair-wise matching cost between ground truth t_i and a prediction with index $\sigma(i)$.

The pair-wise matching costs of the proposed model mainly include the following parts: probability loss $\mathcal{L}_{\text{prob}}$, 3-D structure parameters loss $\mathcal{L}_{3d\text{para}}$, operator loss $\mathcal{L}_{\text{operator}}$, range direction loss $\mathcal{L}_{\text{range}}$, and cylinder IOU loss (CI loss) $\mathcal{L}_{\text{IOU}}$. We define the loss function of our proposed model as follows:

$$\mathcal{L}_{\text{match}} = \lambda_1 \mathcal{L}_{\text{prob}} + \lambda_2 \mathcal{L}_{3d\text{para}} + \lambda_3 \mathcal{L}_{\text{operator}} + \lambda_4 \mathcal{L}_{\text{range}} + \lambda_5 \mathcal{L}_{\text{IOU}} \tag{26}$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5 \in \mathbb{R}$ are hyperparameters that represent the weight of each loss part and these losses are normalized by the number of objects inside the batch. $\mathcal{L}_{\text{prob}}$ is defined as $-\log \hat{\mathcal{P}}_{\sigma(i)}(c_i)$, where c_i represents the class between object or Nonobject. $\mathcal{L}_{3d\text{para}}$ is defined as $\|\mathcal{B}_i - \hat{\mathcal{B}}_{\sigma(i)}\|_1$, where $\mathcal{B}_i$ represents the ground-truth 3-D structure parameters. $\mathcal{L}_{\text{operator}}$ is defined as $\|\mathcal{O}_i - \hat{\mathcal{O}}_{\sigma(i)}\|_1$, where $\mathcal{O}_i$ represents the ground-truth FDO parameters. The range direction loss $\mathcal{L}_{\text{range}}$ is defined as $Loss_{\text{CrossEntropy}}(\mathcal{D}_i, \hat{\mathcal{D}}_{\sigma(i)})$, where $\mathcal{D}_i$ represents the ground-truth range direction.

In order to better improve the speed and performance of the proposed model, we introduce an IOU loss. Similar to the literature [59], consider that three different situations: no overlap, containment, and the ratio of the bottom area to the height is different when the center is completely coincident. The IOU loss $\mathcal{L}_{\text{IOU}}$ is defined as follows:

$$\mathcal{L}_{\text{IOU}} = 1 - \text{IOU}_i + \frac{\rho^2(c_i, \hat{c}_i)}{d^2} + \alpha v \tag{27}$$

where IOU represents the cylinder IOU between ground-truth and predicted cylinders, $\rho(c_i, \hat{c}_i)$ represents the Euclidean distance between the centers of ground-truth and predicted cylinders, and d represents the oblique diagonal length of the smallest cylinder containing the predicted and ground-truth

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cylinders. On the one hand, when two cylinders have no overlapping parts, the addition of this item $((\rho^2(c_i, \hat{c}_i))/d^2)$ increases the center point distance measurement and directly optimizes the center distance of the two cylinders. On the other hand, when the two cylinders are in a containment relationship, the addition of this item can make the model converge faster. Only when the center points are completely coincident, will it degenerate into IoU loss. α is the parameter used for tradeoff, which is defined as follows:

$$\alpha = \frac{\upsilon}{(1 - \mathrm{IOU}_i) + \upsilon} \tag{28}$$

where υ is the parameter used to measure the consistency of the aspect ratios, which is defined as follows:

$$\upsilon = \frac{4}{\pi^2}\left(\arctan\frac{s_i}{h_i} - \arctan\frac{\hat{s}_i}{\hat{h}_i}\right)^2 \tag{29}$$

where s_i and $\hat{s}_i$ represent the bottom area of the ground-truth and predicted cylinders, respectively, and h_i and $\hat{h}_i$ represent the height of the ground-truth and predicted cylinders, respectively. The effect of the item $\alpha\upsilon$ is to add the relative ratio of bottom area and height of the cylinder, which is used to punish the result of the inconsistency between the predicted cylinder shape and the ground truth. The addition of $\mathcal{L}_{\mathrm{IOU}}$ can make our model to have faster convergence speed and better performance than the original loss function without this item.

After obtaining the optimal permutation $\hat{\sigma}$ between the prediction and ground truth, the train loss is similar to $\mathcal{L}_{\mathrm{match}}$, but only use the first M of the predicted objects and the ground-truth objects within $\hat{\sigma}$, where M is the length of ground-truth objects in a batch, and the train loss is defined as follows:

$$\mathcal{L}_{\mathrm{train}} = \mathcal{L}_{\mathrm{match}}\left(t, \hat{t}_{\hat{\sigma}}\right). \tag{30}$$

III. Experiments and Results

In this section, we first introduce the SAR images dataset of oil tank and the setup of our experiments. Then, we analyze the three factors that influence the performance of our proposed model to set up the parameters. Finally, we evaluate our model with ablation experiments, compare experiments, and visualize the model attention maps and results.

A. Dataset Description and Preprocess

For our experiments, the SAR datasets are the L1B SAR images obtained from RADARSAT-2, TerraSAR-X, and GF-3. The detailed information on the datasets is presented in Table I. The dataset includes 21 scenes SAR images. Each scene image corresponds to one flight observation by satellite. Some scenes with different incidence angles are shown in Fig. 8. The labels of dataset include the 3-D structural parameters (location, radius, and height) and the FDO (position of the two strong scattering centers) of every oil tank objects in a single SAR image. In addition, the incidence angle of each scene image and the range direction from near to far is also provided.

We first converted Level 1B SAR images into single-channel grayscale images in tiff format. Then, we copied this channel and converted the single-channel image into the three-channel image to input our proposed model. We selected the area of SAR images that contain oil tank targets and cropped them to the shape of 512*512 pixels. In order to meet the conditions of the SAR image application problem, we did not do any data argumentation to use the known observation conditions. We randomly selected 20% of the dataset for training for our experiments, and the remaining 80% were for testing. A detailed experimental setup is presented in Section III-B.

TABLE I
Detailed Information of RADARSAT-2 Which Our Primary Datasets Are Obtained From

Datasets Information	Parameter
Satellite	Radarsat-2
Sensor	SAR SB
Product Type	SGF
Range Pixel Spacing (m)	1.562500
Azimuth Pixel Spacing (m)	1.562500
Slant Range or Ground Range	Ground Range
Process Level	L1B

B. Experimental Setup

We trained and evaluated our model on an AMD Ryzen 7 5800X 8-Core Processor with 32-GB RAM and an NVIDIA GeForce RTX 3090 with 24-GB RAM. The experimental environment used Pytorch 1.7 deep learning framework with cuda 11.0 and cudnn 8.0 GPU driver. We optimized the model by using the AdamW optimizer [60] setting the initial transformer's learning rate to $2e^{-4}$, the backbone's to $2e^{-5}$, and weight decay to $2e^{-4}$ with a batch size of 8 and employed the loss function described as Section II.6. For fairness, all of our experiments were trained 300 epochs and we dropped the learning rate to 0.1 of initial learning rate at the 200th epoch. In addition, we did not use other training tricks. We evaluated the performance of every methods relying on average precision (AP) and average recall (AR) at cylinder IOU = 0.50:0.05:0.95 and mean offset as follows:

$$\begin{aligned}\text{Average Precision(AP)} &= \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \\ &= \frac{\mathrm{TP}}{\text{All Dectecions}}\end{aligned} \tag{31}$$

$$\begin{aligned}\text{Average Recall(AR)} &= \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \\ &= \frac{\mathrm{TP}}{\text{All Groud Truths}}\end{aligned} \tag{32}$$

$$\begin{aligned}\text{Mean Offset} = \text{Mean}\big(\|&\text{Predictions}\left(\hat{x}, \hat{y}, \hat{r}, \hat{h}\right) \\ &-\text{Groud Truths}(x, y, r, h)\|_1\big)\end{aligned} \tag{33}$$

where true positive (TP) is the number of predictions that cylinder IOU > $\mathrm{IOU}_{\mathrm{threshhold}}$, false positive (FP) is the number of predictions that cylinder IOU < $\mathrm{IOU}_{\mathrm{threshhold}}$, false negative (FN) is the number of ground truths that are not detected, and N is the number of predicted objects in one SAR image.

As for the other default hyperparameters of our experiments, we used resnet50 as a backbone network, sine as the type of positional embedding, 6 as the number of encoding and

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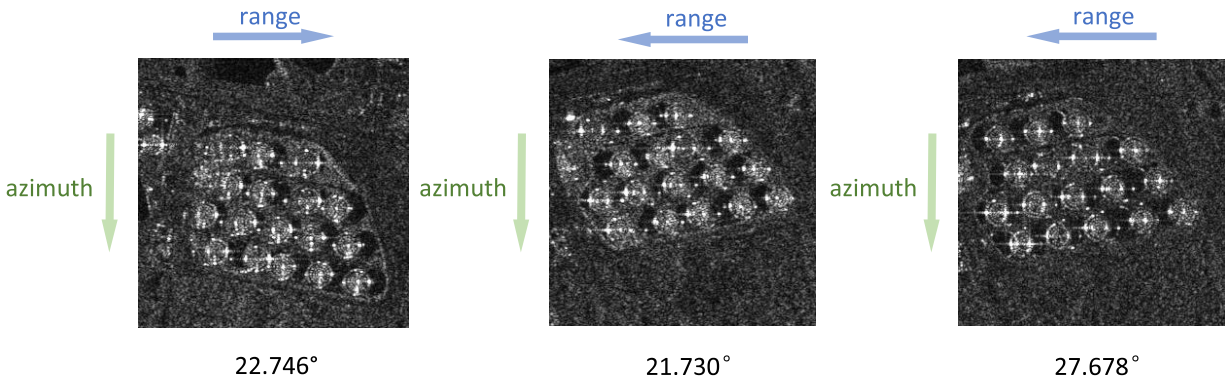


Fig. 8. Some SAR images scenes with different observation conditions. The range, azimuth, and incidence information are provided.

decoder layers in transformer, 256 as the size of embeddings (dimension of the transformer), 1024 as the intermediate size of the feedforward layers in the transformer blocks, 0.1 as the dropout applied in the transformer, 8 as the number of attention heads inside the transformer's attention, 300 as the number of query slots, and 4 as the reference points in encoder and decoder layers. We also used the iterative bounding box refinement proposed in the literature [39].

C. Parameters Setup

In the experiments, we study the influence of three parameters in our proposed model that are depth size, learning rate, and number of attention heads to set up the optimal parameters. In the tables showing the results, we use the three evaluation indicators mentioned above, including AP, AR, and mean offset. The results of AP and AR are shown as percentages, and the unit of mean offset is pixel.

1) Depth Size: We first studied the influence of depth size that is defined as the number of encoder and decoder layers in transformer. We investigated the depth size of 4, 5, 6, 7, and 8, while other hyperparameters were set as described in Section III-B. The results of compared experiments are shown in Table II. At the same time, we also drew a line chart of AP as in Fig. 9 to better show the different results caused by different depth sizes. The data of AP are in the form of percentage, and the data of mean offset represent the mean pixel offset between ground truths and predictions. According to the results, we observe that properly increasing the network depth will improve the performance of the model, but when it increases too much, it will cause overfitting, and the performance of the test set will decrease instead. By comparing the model parameters of different depth sizes that are shown in Table II, we analyze that the total model parameters affect the difficulty during model training increase as well, which is the main reason of overfit. According to the experimental results, when the encoder and decoder blocks' depth is set to 6, the model can achieve the best results. However, when the depth exceeds 6, as the number of the encoder and decoder blocks depth increases, the effect of the total model parameters increases obviously and ultimately reduces the performance of the model.

2) Learning Rate: In this section, we studied the influence of learning rate on model performance. We investigated the

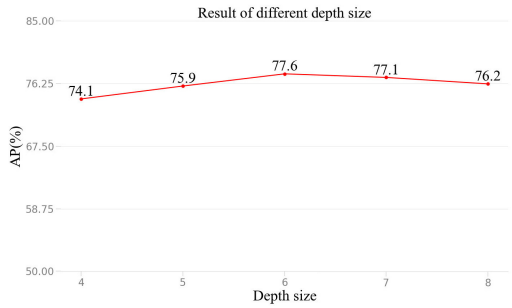


Fig. 9. Results of AP under different depth sizes.

initial model learning rate to $1e^{-4}$, $2e^{-4}$, $4e^{-4}$, and $5e^{-4}$ and the backbone's to $1e^{-5}$, $2e^{-5}$, $4e^{-5}$, and $5e^{-5}$, while other hyperparameters are setting as described in Section III-B. Note that when the initial model learning rate is $1e^{-3}$, the network cannot converge, and when the learning rate is less than $1e^{-4}$, the convergence speed of the network is very slow. We also dropped the learning rate to 0.1 of initial learning rate at the 200th epoch. The results of compared experiments are shown in Table III. We also drew a line chart of AP as in Fig. 10 to show the different results caused by different learning rates. The learning rate affects the convergence speed and performance of the model. From the results, we can see that too high learning rate will cause the model to achieve poor test performance because the model will not continue to converge, while too low learning rate will make the network unable to achieve high AP results within a fixed training period of 300 epochs, and ultimately, the initial model learning rate of $2e^{-4}$ achieves the highest AP result.

3) Number of Attention Heads: In this section, we studied the influence of different numbers of attention heads on model performance. We investigated the number of attention heads of 4, 8, and 16, while other hyperparameters were set as described in Section III-B. The results of compared experiments are shown in Table IV. We also drew a line chart of AP as in Fig. 11 to show the different results caused by different numbers of attention heads. The transformer networks we used rely on the multihead attention mechanism and cross-attention mechanism, which runs multiple attention layers ("heads") in parallel, connecting their outputs. Mul-

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TABLE II
RESULTS OF COMPARED EXPERIMENTS WITH DIFFERENT DEPTH SIZES

Method	Depth_{enc}	Depth_{dec}	parameters	$\text{AP}_{0.50:0.95}(\%)$	$\text{AR}_{0.50:0.95}(\%)$	Mean Offset (pixel)
Exp-A1	4	4	36.37M	74.1	81.7	1.1136
Exp-A2	5	5	38.18M	75.9	84.1	0.8576
Exp-A3	6	6	40.58M	**77.6**	83.9	**0.7424**
Exp-A4	7	7	41.80M	77.1	83.7	0.7643
Exp-A5	8	8	43.61M	76.2	84.1	0.7808

TABLE III
RESULTS OF COMPARED EXPERIMENTS WITH DIFFERENT LEARNING RATES

Method	lr_{init}	$\text{lr}_{backbone}$	parameters	$\text{AP}_{0.50:0.95}(\%)$	$\text{AR}_{0.50:0.95}(\%)$	Mean Offset (pixel)
Exp-B1	1e-4	1e-5	40.58M	74.3	82.7	1.216
Exp-B2	2e-4	2e-5	40.58M	**77.6**	**83.9**	**0.7424**
Exp-B3	4e-4	4e-5	40.58M	71.4	83.6	1.2866
Exp-B4	5e-4	5e-5	40.58M	70.0	81.7	1.3312

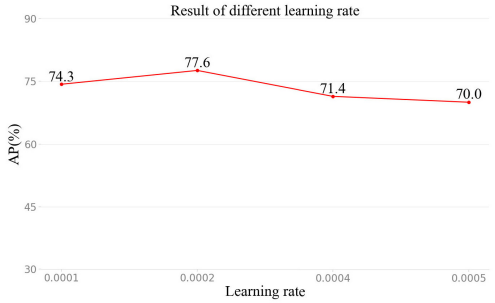


Fig. 10. Results of AP under different learning rates.

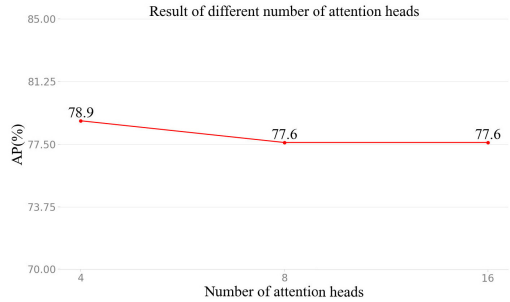


Fig. 11. Results of AP under different numbers of attention heads.

tihead attention, without increasing time complexity, makes it easier for the transformer network to learn more abundant feature information in multiple-independent feature spaces in lower dimensions. It can be seen from the results that it is not the more attention heads, the better performance of the model. Although increasing the number of attention heads can form multiple subspaces to let the model pay attention to different aspects of information, it will also cause redundancy of attention and the model parameters increase, which will increase the difficulty of network training and reduce the performance of the model. In our experiments, the attention heads of four achieve the highest AP result.

D. Evaluation Results

In this section, we first introduce the ablation studies for evaluation. Next, we introduce the compared experiments between our models and other SOTA methods to prove the superiority and effectiveness of our methods. Then, we visualize the prediction results of our proposed model. We will discuss the results of all the experiments and the effectiveness of our proposed model in Section IV.

1) Ablation Studies Results: To evaluate every modules' impact on the proposed model performance, we conduct several ablation studies of PT, CI loss, and FDO. Next, we introduce the designs and results of ablation experiments.

We regard Exp-D1 as a baseline, which removed the CI Loss, PT, and FDO compared with our model. For Exp-D2, we added the CI Loss on the basis of the baseline. For Exp-D3, we added PT compared with Exp-D2. The results of compared experiments are shown in Table V. According to the results, we observe that all the three modules have achieved positive effects on evaluation metrics, which proved the effectiveness of our proposed model. We will discuss more in Section IV.

2) Compared Experiments Results: Table VI shows the comparisons of our method and other SOTA methods. Our approach uses deformable-DETR as the base architecture, and we test the performance of deformable-DETR with cylinder IOU. In addition, we test other SOTA CNN-based methods using MMDectection toolbox [61] on our SAR dataset. For detectors [62], we use ResNet-50 as backbone. For hybrid task cascade [63], we use HRNet-W40 and ResNeXt-101 as backbone. For cascade mask R-CNN [64], we use ResNet-101 and ResNeXt-101 as backbone. Notice that cascade mask R-CNN, detectors, and hybrid task cascade are CNN-based methods, which achieves the best results in object detection. Deformable-DETR is a transformer-based method, which benefits from the multiscale self-attention mechanism and manages AP of 3.3% gain over the best method of CNN-based methods in our experiments. Our model benefits from the three improved modules we proposed, which are PT, CI Loss, and FDO, and manages AP of 4.2% gain over the deformable-DETR in our experiments. The results of AR and mean offset also prove the superiority and effectiveness of our method.

3) Visualization Results: In order to verify the effectiveness of our proposed model more concretely, we visualize the attention sampling points and attention weights in the last layer of model's encoder and decoder. We list several samples' results of test sets as in Fig. 12, which include encoder attention sampling points map, decoder attention sampling points map,

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TABLE IV
RESULTS OF COMPARED EXPERIMENTS WITH DIFFERENT NUMBERS OF ATTENTION HEADS

Method	nheads	parameters	$\mathbf{AP}_{0.50:0.95}(\%)$	$\mathbf{AR}_{0.50:0.95}(\%)$	Mean Offset (pixel)
Exp-C1	4	39.98M	**78.9**	83.9	**0.7296**
Exp-C2	8	40.58M	77.6	83.9	0.7424
Exp-C3	16	41.77M	77.6	83.9	0.7296

TABLE V
RESULTS OF ABLATION STUDIES OF PT, CI LOSS, AND FDO

Method	CI Loss	PT	FDO	$\mathbf{AP}_{0.50:0.95}(\%)$	$\mathbf{AR}_{0.50:0.95}(\%)$	Mean Offset (pixel)
Exp-D1				60.8	69.8	2.496
Exp-D2	$\surd$			74.7	82.6	0.8576
Exp-D3	$\surd$	$\surd$		75.8	**84.4**	0.7808
Ours	$\surd$	$\surd$	$\surd$	**77.6**	83.9	**0.7424**

TABLE VI
COMPARISONS OF OUR APPROACH AND OTHER SOTA METHODS ON OUR SAR DATASET

Method	backbone	parameters	$\mathbf{AP}_{0.50:0.95}(\%)$	$\mathbf{AR}_{0.50:0.95}(\%)$	Mean Offset (pixel)
Cascade Mask R-CNN [64]	ResNet-101	87.92M	67.3	74.6	1.7143
Cascade Mask R-CNN	ResNeXt-101	126.65M	68.6	75.5	2.5057
Detectors [62]	ResNet-50	131.2M	53.1	63.4	1.4021
HTC [63]	ResNeXt-101	131.79M	65.3	73.5	1.9849
HTC	HRNet-W40	90.92M	71.4	78.1	2.4895
D-Detr [39]	ResNet-50	40.49M	74.7	82.6	0.8576
Ours	ResNet-50	**39.98M**	**78.9**	**83.9**	**0.7296**

reference points map, and 2-D result map. Encoder and decoder attention sampling points map combines the sampling points and attention weights of different resolution feature maps and different attention heads. Each sampling point is drawn as a filled circle, whose depth of color represents the attention weight of the detected object. We also drew the reference points map, which represents the center point of precision. We drew the bottom and top circles of the cylinder oil tank on the SAR image in the 2-D result map. The yellow circle represents the top circle, while the blue one represents the bottom circle. From the visualization results, we observe that our models can precisely focus on all the objects in a single input image. The object centers and the bottom and top circles predicted by our model are also precisely located, which proves the effectiveness of our proposed method.

Finally, we show the final 3-D detection results of our model in test sets. Given one image in test sets, we obtained the 3-D structure parameters of all the oil tank objects through our model. Then, we generated the 3-D models on the input SAR image. Some samples of rendering maps are shown in Figs. 13 and 14. From the visualization results of rendering results, our method can accurately extract the 3-D location and structure parameters of oil tank and realize end-to-end 3-D detection of oil tank from single SAR image.

IV. DISCUSSION

In specific experiments, we first studied the influence of different model's hyperparameters on the results. Similar to most object detection problems, AP is the most important evaluation metric which we focus on. The best results we obtained when setting depth size to 6, learning rate to $2e^{-4}$, and the number of attention heads to 4. For each type of parameter, we changed the value of it while keeping the other parameters the same for fairness. We have also discussed the relevant results in Section III-C.

In our proposed model, we put forward PT, CI Loss, and FDO modules on the basis of baseline model that is described in Section III-D1. We conduct several ablation studies to evaluate the effectiveness of these modules. In Exp-D2, we added CI Loss in the loss function compared with baseline. For results, it manages AP of 74.7%. Compared with the baseline that manages AP of 60.8%, it can greatly improve the convergence speed and convergence result, which avoids the situation that with the decrease of predicted parameters' mean offset, but the AP does not decrease. In Exp-D3, we added PT in the input of transformer network compared with Exp-D2. It manages AP of 75.8%, which proves that PT allows the model to better adjust the prediction results according to the provided incidence angle information and improve the precision of 3-D structural parameter prediction. The best results are attained in our proposed model managing AP of 77.6%. We added FDO modules compared with Exp-D3. We observe that FDO modules allow the model to focus on the position, which reflects the 3-D information in the SAR image.

Since our proposed model uses a multiscaled transformer network, it mainly relies on multiscale attention mechanism. To evaluate the superiority and effectiveness of transformer architecture, we conduct some compared experiments between transformer-based methods and CNN-based methods. From the results shown in Table VI, the best result of CNN-based methods is HTC ($\mathrm{HRNet}-W40$), which manages AP of 71.4%. The transformer-based methods deformable-Detr ($\mathrm{ResNet}-50$) manages AP of 74.7%, which achieves better results using fewer parameters. We analyze the SAR

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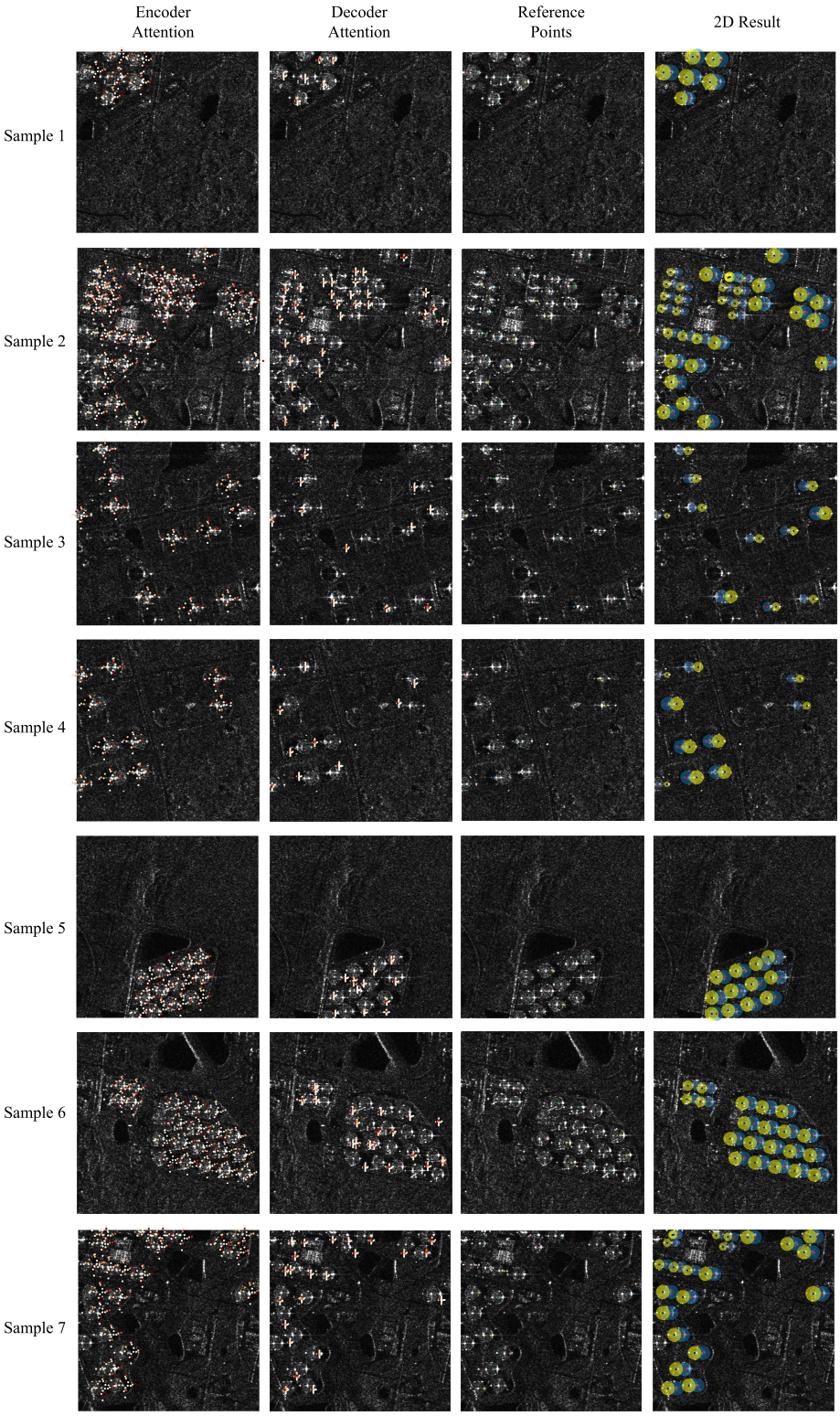


Fig. 12. Visualizations of the attention sampling points and attention weights in the last layer of model's encoder and decoder. From left to right are encoder attention sampling points map, decoder attention sampling points map, reference points map, and 2-D result map. The darker color represents the greater weight of attention, that is, the more focused part of the multiscale transformer network. The reference points are the sampling point finally selected by the model after training. These reference points indicate the centers of all targets in the image predicted by the model. All of other parameters predicted by the model are calculated relative to the reference points.

dataset and observe that the oil tank targets in the SAR image consist of a lot of discrete and discontinuous strong scattering centers. In order to better extract the oil tank target with the above features, the network needs to have a larger receptive field. CNN-based methods have a good effect in extracting the features of local regions, but the receptive

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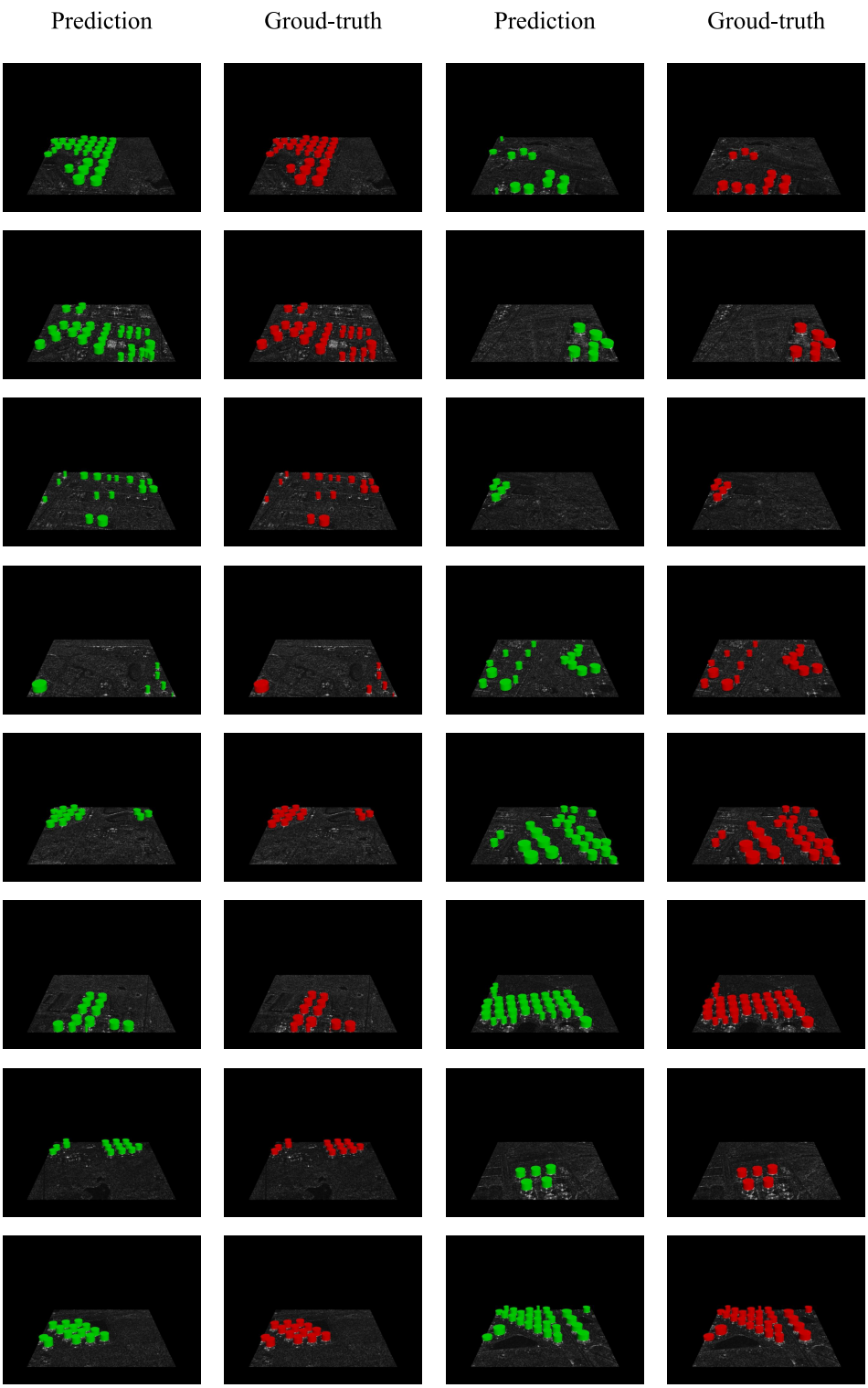


Fig. 13. 3-D detection rendering results (Part 1) of our model in test sets. The first and third columns are prediction results and the second and fourth columns are ground-truth results.

field of the network is limited, which affects the effect of extracting oil tank targets. In addition, since the characteristics of SAR images are reflected in the intensity value of each pixel, that is, the intensity of the scattering point, there is a lot of interference in the SAR image containing the oil tank target. We observe that real oil tank targets and interfering background targets are both existing in a single SAR image, which greatly increases the false detection rate of CNN-based methods because of the limited receptive field. We show the real oil tank targets and interfering targets in

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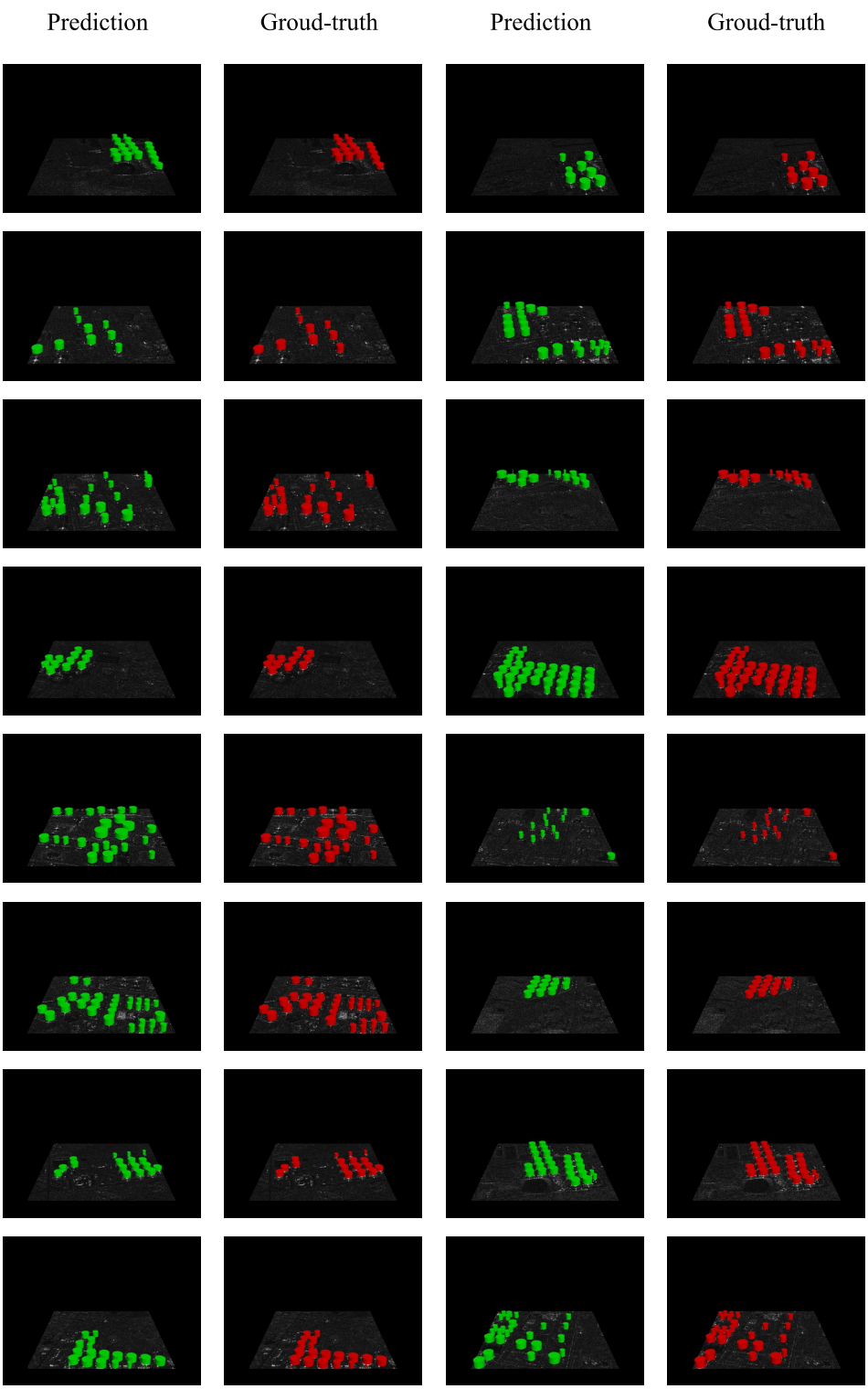


Fig. 14. 3-D detection rendering results (Part 2) of our model in test sets. The first and third columns are prediction results and the second and fourth columns are ground-truth results.

the background in Fig. 15. Benefitted from the multiscale self-attention mechanism, the transformer-based methods can integrate long-range image features to obtain the attention weight of the whole image, which helps the model to find the similarity information between the real oil tank targets and distinguish the background interfering targets and the real oil tank targets. The CNN-based methods are limited by the receptive field and pay more attention to local features. Thus, they are not as good as the transformer-based methods in

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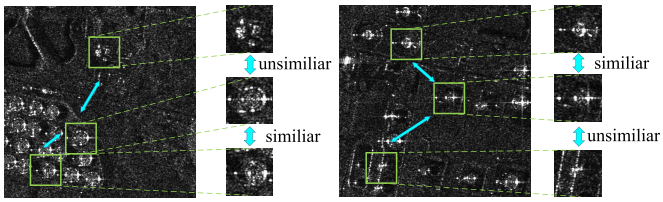


Fig. 15. Real oil tank targets and interfering targets in the background of two-sample SAR images. Our proposed model can reasoning and discriminating the similarity of real oil tank targets and interfering targets.

reasoning and discriminating the similarity of candidate targets in the whole image.

Benefitted from the strong interpretability of the transformer network, we visualized the sampling points map and attention weights of the multiscale attention module in Section III-D2. According to the attention map of encoder and decoder, our model is able to locate all the oil tanks in single SAR image precisely. At the same time, our model can adopt different resolution sampling points and attention weights to different scaled objects. The results in reference points map and 2-D result map reflect that the model can accurately locate the objects. According to the range direction, we can accurately locate the bottom circle and top circle of oil tank objects. Finally, to better view the 3-D detection results of our model, we also generate 3-D models on SAR images based on the predicted structural parameters. The rendering map is shown in Fig. 13. We can concretely observe that the 3-D structure predicted by our model is close to the ground-truth value, and our method can accurately realize the end-to-end 3-D detection of oil tank from a single SAR image.

V. Conclusion

In this article, we proposed an end-to-end method with transformer for 3-D detection of oil tank from a single SAR image. First, we analyze the imaging mechanism and characteristics of the SAR image in detail. Since the past application of deep learning in the field of SAR image object detection did not consider the unique characteristics of SAR images, we combine the unique characteristics of SAR image by using transformer networks. The CNN-based model uses convolution to extract the features of SAR image, and the transformer network is more suitable for in-depth detection result inference. Thus, we used the CNN as the backbone and transformer network as an inference module. We input the incidence angle information as *a priori* information into the model and consider utilizing scattering information as FDO to improve the performance. Through ablation experiments on a large number of SAR image datasets with different incidence angles, our model achieves the AP of 77.6% compared with 60.8% of baseline, which proved the effectiveness of our proposed modules. We also conduct compared experiments between CNN-based methods and transformer-based methods, the results of which proved the superiority of transformer-based methods. We regard the oil tank object as a standard cylinder, so we propose cylinder IOU to better evaluate the predicted parameters' precision of our model. In addition, we also study the influence of different parameters in our model which contain depth size, learning rate, and the number of attention heads to set up the optimal parameters. Finally, we discussed the reasons why our method is superior and effective. Through these studies of our article, we explore a way to combine the unique characteristics of SAR image and deep learning. Our model utilizes the traditional features of SAR images and combines the advantages of deep learning in the field of long-term and large-scale object detection. We proved that the oil tank object's 3-D information can be extracted from a single SAR image. The transformer model makes this method a success. Our model provides a new way to detect typical targets in SAR images combining the unique characteristics of SAR image and transformer model, which can inspire more applications in the field of SAR targets interpretation.

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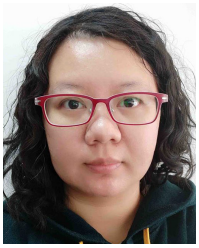
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